

UNESCO Youth Hackathon 2026

Project Proposal

TrustLens AI: Think Before You Trust

Team Name: TrustLens

Category: Applications / Websites

Country: Philippines

1. Executive Summary

Imagine a world where every image, video, or news story you see online could be artificially generated, manipulated, or completely fabricated, yet almost impossible to distinguish from reality. As generative Artificial Intelligence (AI) rapidly reshapes the digital landscape, misinformation now spreads faster, farther, and more convincingly than ever before. Deepfakes, AI-generated images, fabricated videos, and hallucinated AI responses have become emerging global threats that undermine public trust, informed decision-making, and democratic participation (UNESCO, 2023; World Economic Forum, 2025). While AI has revolutionized access to information, it has also created an urgent need to strengthen Media and Information Literacy (MIL) and equip young people with the critical thinking skills necessary to navigate an increasingly complex information ecosystem (Walter et al., 2020).

TrustLens AI: Think Before You Trust is an AI-powered web and mobile platform that transforms Media and Information Literacy into an interactive, real-time experience. Rather than functioning as a conventional fact-checking application, TrustLens AI empowers users to understand why information can be trusted through explainable AI, transparent Trust Scores, source verification, AI-generated content detection, and contextual analysis. By combining artificial intelligence with behavioral science and education, the platform encourages users to verify information before believing or sharing it, fostering lifelong habits of critical thinking and responsible digital engagement.

The platform introduces two signature innovations that distinguish it from existing solutions. Pause Before Sharing encourages users to stop and review evidence before reposting questionable content, while Trust Replay visually traces how misinformation spreads across digital platforms, helping users understand its origins, amplification, and societal impact. Complemented by gamified learning modules, achievement badges, and community-driven challenges, TrustLens AI transforms digital verification into an engaging educational journey that empowers youth to become active defenders of truthful and responsible information.

More than a technology solution, TrustLens AI is a movement to rebuild digital trust. By empowering young people with the knowledge, skills, and confidence to critically evaluate

information in the age of AI, the platform seeks to cultivate a generation of informed, ethical, and digitally resilient citizens capable of strengthening democratic participation, promoting responsible AI use, and creating a safer, more trustworthy digital future.

2. TECHNICAL ARCHITECTURE

Core Components

Component	Purpose
TrustLens AI Engine	Performs AI-powered credibility assessment and content analysis.
Trust Score Analyzer	Calculates content reliability using source credibility, contextual analysis, and AI-generated content detection.
Explainable AI Module	Generates transparent explanations showing why content is trustworthy or misleading.
Trust Replay Engine	Visualizes how misinformation spreads across digital platforms through interactive timelines.
Pause Before Sharing Module	Encourages users to verify suspicious content before reposting through behavioral nudges.
Learning Management Module	Delivers interactive Media and Information Literacy (MIL) lessons, quizzes, badges, and progress tracking.
Administration Dashboard	Manages users, educational content, analytics, moderation, and AI governance.

Supporting Technologies

Layer	Technology
Frontend	React 19 + TypeScript + Tailwind CSS
Backend	Node.js + Express.js
Database	PostgreSQL
Authentication	Firebase Authentication / OAuth 2.0
AI Engine	OpenAI GPT-4.1 / DeepSeek Chat (Model Agnostic)
AI Content Detection	Hive Moderation API / OpenAI Moderation API
Cloud Storage	Firebase Storage
Hosting	Vercel + Supabase
Analytics	Google Analytics 4

System Workflow

1. User registers and creates a secure account.
2. User pastes a URL, uploads an image, or submits text for verification.
3. The AI Engine analyzes the content using source verification, contextual analysis, and AI-generated content detection.
4. TrustLens generates a transparent Trust Score with explainable reasoning.

5. If content is potentially misleading, the **Pause Before Sharing** feature prompts users to review evidence before reposting.
6. Users explore **Trust Replay** to understand how misinformation spreads.
7. Learning modules recommend relevant Media and Information Literacy lessons based on verification results.
8. Administrators monitor analytics, educational engagement, and system performance through the governance dashboard.

3. AI APPROACH & MODEL SELECTION

TrustLens AI employs an **Explainable Retrieval-Augmented Prompting (ERAP)** approach that combines prompt engineering, source verification, explainable artificial intelligence, and behavioral interventions instead of relying solely on conventional AI-generated classifications.

AI Features

1. Trust Score Assessment — Evaluates the credibility of online content using multiple verification indicators.
2. Explainable AI — Provides human-readable explanations rather than binary "true" or "false" decisions.
3. AI-Generated Content Detection — Identifies AI-generated text and images through external detection services.
4. Trust Replay — Visualizes the propagation of misinformation across digital platforms.
5. Personalized Learning Recommendations — Suggests MIL lessons based on users' verification history.

Key Design Decisions

Design Choice	Rationale
Explainable AI	Builds transparency and user trust by explaining every assessment.
Multi-factor Trust Score	Avoids oversimplified true-or-false classifications.
Behavioral Nudging	Encourages responsible online behavior before sharing content.
Human-Centered AI	Supports critical thinking rather than replacing user judgment.
Model-Agnostic Architecture	Allows seamless switching between OpenAI, DeepSeek, Gemini, or future AI providers.

AI Techniques

- Prompt engineering with structured verification prompts
- Explainable AI (XAI)
- Source credibility verification
- Context-aware reasoning
- Reverse image search integration
- AI-generated content detection
- Behavioral intervention prompts
- Educational recommendation engine

Selected Model

Primary Model: OpenAI GPT-4.1

Alternative Model: DeepSeek Chat

Reason for Selection

- Strong reasoning capability
- Excellent natural language understanding
- Explainable responses
- API scalability
- Cost-effective deployment
- Provider-independent architecture

SECTION 4: DATA STRATEGY & ETHICS

Data Collected

TrustLens AI collects only information necessary to provide verification and educational services.

- User profile information
- Submitted URLs
- Uploaded images
- Verification requests
- Learning progress
- Quiz scores
- Badge achievements
- Anonymous analytics
- User consent records

Data Protection Measures

Principle	Implementation
Data Minimization	Only content submitted for verification is processed.
User Consent	AI features require explicit user agreement.
Encryption	HTTPS and encrypted cloud storage protect user data.
Transparency	Explainable AI reveals how verification decisions are generated.
Privacy	Personal information is never shared with third parties without consent.

Ethical Principles

TrustLens AI follows internationally recognized AI ethics principles:

- Transparency
- Fairness
- Accountability
- Privacy
- Inclusiveness
- Human oversight
- Responsible AI

- Media and Information Literacy

Bias Mitigation

Potential risks include:

- AI hallucinations
- Political bias
- Cultural bias
- Language limitations

Mitigation strategies include:

- Multi-source verification
- Confidence indicators
- Human review mechanisms
- Continuous model evaluation
- Explainable AI outputs
- Regular bias monitoring

Data Lifecycle

1. User submits digital content.
2. User grants explicit consent for AI processing.
3. AI analyzes content and generates a Trust Score.
4. Verification results and explanations are displayed.
5. Learning recommendations are generated.
6. Anonymous analytics are stored securely.
7. Users may delete their data or withdraw consent at any time.

SECTION 5: DEVELOPMENT MILESTONES (AGILE ROADMAP)

Sprint	Major Deliverables
Sprint 1	User authentication, onboarding, Trust Score prototype, database setup, UI/UX design
Sprint 2	AI verification engine, Explainable AI module, source verification integration
Sprint 3	Trust Replay visualization, Pause Before Sharing, learning modules
Sprint 4	Gamification, quizzes, badges, analytics dashboard
Sprint 5	Testing, security, accessibility improvements, deployment

Final Prototype Outcomes

- Responsive web application
- Mobile-ready interface
- AI-powered Trust Score system
- Explainable AI verification
- Trust Replay visualization
- Pause Before Sharing behavioral intervention
- Interactive Media and Information Literacy lessons

- Gamification system
- Analytics dashboard
- Administrator governance portal

SECTION 6: SCALABILITY & REGIONAL RESILIENCE

Scaling Potential

TrustLens AI is designed for global deployment through:

1. Cloud-native architecture supporting millions of users.
2. Modular AI services enabling future model replacement.
3. Multi-language support for international expansion.
4. School and university integration.
5. API interoperability with fact-checking organizations.
6. Containerized deployment using Docker and Kubernetes.
7. Regional hosting for data sovereignty compliance.
8. Mobile-first architecture optimized for low-bandwidth environments.

Technical Constraints & Mitigation

Constraint	Mitigation
AI hallucinations	Multi-source verification and explainable AI outputs
External AI dependency	Model-agnostic architecture supporting multiple providers
API latency	Asynchronous processing with background queues
Language limitations	Multilingual prompt engineering and localization
Emerging misinformation patterns	Continuous AI model updates and human moderation
Privacy regulations	Compliance with GDPR and regional data protection standards
Scalability challenges	Cloud auto-scaling and distributed infrastructure
User trust	Transparent explanations and confidence scoring

References:

- UNESCO. (2023). Guidance for generative AI in education and research. UNESCO.
<https://unesdoc.unesco.org/ark:/48223/pf0000386693>
- Walter, N., Cohen, J., Holbert, R. L., & Morag, Y. (2020). Fact-checking: A meta-analysis of what works and for whom. *Political Communication*, 37(3), 350–375.
<https://doi.org/10.1080/10584609.2019.1668894>
- World Economic Forum. (2025). The Global Risks Report 2025 (20th ed.). World Economic Forum.